

Sanithu D. Heengama

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EDUCATION

Brown University

Sc.B. in Computer Engineering & Applied Mathematics, GPA: 3.8/4.0

Providence, RI

Expected May 2027

Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, Computational Mathematics, Quantum Mechanics

RESEARCH EXPERIENCE

3D from Multi-View and Sensors, Brown University (LEMS)

Aug 2025 – Present

Advisors: Prof. Benjamin Kimia, Chiang-Heng Chien (Ph.D. Student)

- Build a multi-view 3D reconstruction pipeline: camera calibration, epipolar geometry, essential/fundamental matrix estimation, triangulation, and optimization for pose recovery.

Musculoskeletal Joint & Implant Tracking — Research & SWE Intern

July 2026 – Present

Bioengineering Laboratory, Orthopaedic Research, Brown University Health | Mentor: Prof. J.J. Trey Crisco III

- Extended open-source 3D Slicer tools for markerless skeletal tracking from biplane imaging, applying multi-view reconstruction to a real clinical pose-estimation system.

Radiance Fields for 3D Computer Vision and AI, Brown University

May 2025 – Dec 2025

Advisors: Prof. Srinath Sridhar, Angela Xing

- Implemented deformable 3D Gaussian splatting reconstruction in PyTorch for ground-truth particle position/velocity; built a visuotactile manipulation dataset with a custom UMI gripper.

AI Physics Game — RL & Physics-Informed Neural Nets, Brown University

Sept 2025 – Jan 2026

Advisor: Prof. Rick Gaitskell

- Designed reinforcement-learning agents and a multi-agent gravitational-interaction environment for stable orbital control.

ADDITIONAL RESEARCH EXPERIENCE

- Esophageal Manometry Redesign, Brown University & RISD (Sept 2025 – May 2026) — co-led embedded electronics/software for a patient-facing diagnostic device.
- Reaction-Diffusion Pattern Formation Theory, Applied Math DRP, Brown University (Feb – May 2025) — modeled pattern formation; presented at DRP 2025 Workshop.
- Engineering Research Intern, Dexwet Holding Corporation, Vienna, Austria (June – Aug 2024) — developed 9 IoT air-quality prototypes (CAD, MATLAB).

PUBLICATIONS

- Heengama, S. (2026). Word-Specific Repetition Priming Drives Character-Level Language Intrusion in Quantized Reasoning Models. <https://doi.org/10.5281/zenodo.22108468>
- Heengama, S. "A Cost-Effective IoT-Based Intelligent Indoor Air Quality Monitor." Journal of Emerging Investigators, JEI-23-024R4 (first author, 2023). doi.org/10.59720/23-024

AWARDS, TEACHING & TRAINING

- 3× UTRA/SPRINT Fellowship, Brown University; HKIMO Global Final — World Star & Bronze Medal; 2× International Scholar with Honors; 2× Pearson Edexcel Excellence Award.
- Undergraduate Teaching Assistant, ENGN 40 Dynamics & Vibrations and ENGN 32 Intro to Engineering: Design, Brown University.
- LPS Qubit Collaboratory (LQC) Quantum Computing Program — circuits, algorithms, error correction, qubit technologies (Summer 2025).

SKILLS

Languages: Python, C++, C, Java, MATLAB, Verilog

ML / CV: PyTorch, scikit-learn, NumPy, SciPy, CUDA, 3D Reconstruction, Gaussian Splatting

Hardware / EE: ROS2, Cadence, Fusion 360, SolidWorks, Qiskit